

Cheng-Hsiang Chiu

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QUALIFICATION

- Ph.D. with expertise in parallel and heterogeneous computing, scheduling algorithms, and runtimes
- Hands-on experience building parallel libraries, GPU workloads, AI infrastructure, and LLM applications
- Proven ability to collaborate with cross-disciplinary teams and translate research into efficient solutions

EDUCATION

- **University of Wisconsin-Madison** Madison, Wisconsin, USA
Ph.D. in Electrical and Computer Engineering Aug. 2023 – Aug. 2025
- **École Polytechnique Fédérale de Lausanne** Lausanne, Switzerland
M.S. in Computer Science Sep. 2013 – Feb. 2016
- **National Chiao Tung University** Hsinchu, Taiwan
M.S. in Communications Engineering Sep. 2005 – Aug. 2007
- **National Chung Cheng University** Chiayi, Taiwan
B.S. in Electrical Engineering Sep. 2001 – Jun. 2005

PROJECTS

- **AI-powered applications:** Integrated AI into software applications using APIs and open-source libraries
- **RL Scheduling:** Developed RL scheduling algorithms for work-stealing runtime and CUDA Graph
- **Taskflow:** Developed Pipeflow, AsyncTask, syclflow in Taskflow (<https://taskflow.github.io>).

SELECTED EXPERIENCE

- **Software Intern at Intel** Austin, Texas, USA
Developed the implicit SYCL Graph programming model May 2022 - Aug. 2022
- **Software Intern at Cadence** Austin, Texas, USA
Accelerated the executions of buffer insertion algorithm by 16% May 2021 - Aug. 2021

SELECTED PUBLICATIONS

- **C.H. Chiu**, C. Morchdi, C.C. Chang, C. Yu, Y. Zhou, and T.W. Huang, "Optimizing CUDA Graph Scheduling with Reinforcement Learning - A Case Study in SSTA Propagation," *MLCAD*, 2025
- **C.H. Chiu**, C. Morchdi, Y. Zhou, B. Zhang, C. Chang, and T.W. Huang, "Reinforcement Learning-generated Topological Order for Dynamic Task Graph Scheduling", *HPEC*, 2024.
- C. Morchdi, **C.H. Chiu**, Y. Zhou, and T.W. Huang, "A Resource-efficient Task Scheduling System using Reinforcement Learning," *ASP-DAC*, 2024
- **C.H. Chiu**, Z. Xiong, Z. Guo, T.W. Huang, and Y. Lin, "An Efficient Task-parallel Pipeline Programming Framework," *HPC Asia*, 2024
- **C.H. Chiu**, D.L. Lin, and T.W. Huang, "Programming Dynamic Task Parallelism for Heterogeneous EDA Algorithms," *ICCAD*, 2023

SKILLS

- **Programming Language:** C, C++, Python, HTML, SQL
- **Programming Model:** Taskflow, SYCL, CUDA, OpenMP, MPI, PyTorch
- **AI & LLM Systems:** OpenAI, Llama, Embeddings, Vector Databases, LangChain, Hugging Face, LLMOps
- **Testing & Profiling:** doctest, gprof, perf, Valgrind
- **DevOps & Tooling:** GitHub, CI/CD, Docker, CMake